

Solutions for Analysis I by Terence Tao

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Remark. This document follows the third edition of *Analysis I*.

1 Introduction

No exercises from this chapter.

2 Starting at the beginning: the natural numbers

2.1 The Peano axioms

No exercises from this subchapter.

2.2 Addition

Exercise 2.2.1. *Prove Proposition 2.2.5 (Associativity of addition): For any natural numbers a, b, c , we have $(a + b) + c = a + (b + c)$.*

Solution. We induct on a , keeping b and c fixed.

First, consider the base case $a = 0$. Then we have $(0 + b) + c = 0 + (b + c)$. By Lemma 2.2.2, we obtain $b + c = b + c$ as desired.

Now we assume inductively that $(a + b) + c = a + (b + c)$; we now have to show $((a++) + b) + c = (a++) + (b + c)$ to close the induction. By applying Lemma 2.2.3 twice on the left-hand side, $((a++) + b) + c = ((a + b)++) + c = (a + b + c)++$. Similarly, applying Lemma 2.2.3 on the right-hand side yields $(a++) + (b + c) = (a + b + c)++$. Thus, both sides are equal, and we have closed the induction. $\square$

Exercise 2.2.2. *Prove Lemma 2.2.10: Let a be a positive number. Then there exists exactly one natural number b such that $b++ = a$.*

Solution. We induct on a .

First, consider the base case $a = 1$. By Definition 2.1.3, $1 = 0++$. Suppose there is another natural number b such that $b++ = 1$. However, by Axiom 2.4, if $b++ = 0++ = 1$, then $b = 0$. Thus, there is exactly one natural number b such that $b++ = 1$. Hence, the base case is proven.

Suppose inductively that there exists exactly one natural number b such that $b++ = a$. We now have to show there exists exactly one natural number b' such that $b'++ = a++$. By the inductive hypothesis, $a++ = (b++)++ = b'++$. As per Axiom 2.4, there exists exactly one solution for b' , namely $b' = a$. This closes the induction. $\square$

Exercise 2.2.3. *Prove Proposition 2.2.12 (Basic properties of order of natural*

numbers).

Solution. Let a, b, c be natural numbers. Then

(a) *Reflexivity:* $a \geq a$.

By Lemma 2.2.2, $a + 0 = a$. Thus, by Definition 2.2.11, $a \geq a$.

(b) *Transitivity:* If $a \geq b$ and $b \geq c$, then $a \geq c$.

By Definition 2.2.11, write $a = b + m$, $b = c + n$, where m and n are natural numbers. Thus, $a = (c + n) + m = c + (n + m)$ by associativity of addition. By mathematical induction, it is trivial to prove that $n + m$ is a natural number. Therefore, $a \geq c$. $\square$

(c) *Anti-symmetry:* If $a \geq b$ and $b \geq a$, then $a = b$.

By Definition 2.2.11, if $a \geq b$, then for a natural number m , $a = b + m$. But also if $b \geq a$, then $b = a + n$, where n is a natural number. Thus $a = (a + n) + m = a + (n + m)$. By Lemma 2.2.2 and Proposition 2.2.6 (cancellation law), $0 = n + m$. By Corollary 2.2.9, this results in $m = n = 0$. Thus $a = b + 0 = b$ as required. $\square$

(d) *Preservation of order:* $a \geq b$ if and only if $a + c \geq b + c$.

First, prove $a \geq b \implies a + c \geq b + c$. According to Definition 2.2.11, if $a \geq b$, then $a = b + m$, where m is a natural number. Thus $a + c = (b + m) + c = (b + c) + m$, thus $a + c \geq b + c$.

Conversely, consider $a + c = (b + c) + m$, where m is a natural number. By Proposition 2.2.6 (cancellation law) and the associativity of addition, $a = b + m$, meaning $a \geq b$.

Because $a \geq b \implies a + c \geq b + c$ and $a + c \geq b + c \implies a \geq b$, $a + c \geq b + c \iff a \geq b$. $\square$

(e) $a < b$ if and only if $a++ < b$.

First, consider $a < b \implies a++ \leq b$. According to Definition 2.2.11, if

$a < b$, then $b = a + m$, where m is a natural number. Note that $m \neq 0$ because we would otherwise obtain $a = b$, which is contradictory to the given condition $a < b$. Therefore, by Axiom 2.3, we have $m = n++$, where n is a natural number. Thus, by the definition and commutativity of addition, $b = a + m++ = (a + m)++ = (a++) + m$. It follows that $a \leq b$.

Now consider the converse. From $a++ \leq b$, we obtain $b = a++ + p$, where p is a natural number. Thus, by the definition and commutativity of addition, $b = a++ + p = (a + p)++ = a + p++$. According to Axiom 2.3, 0 is not the successor of any natural number, so $p++ \neq 0$. Hence, $a \leq b$.

Because $a < b \implies a++ \leq b$ and $a++ \leq b \implies a < b$, $a < b \iff a++ \leq b$. $\square$

(f) $a < b$ if and only if $b = a + d$ for some positive number d .

First consider $a < b \implies b = a + d$. From $a < b$ we know $a \geq b$, or $b = a + d$ for a natural number d . For the sake of contradiction, suppose $d = 0$. Therefore $a = b$, which contradicts $a < b$. Hence, d is positive; thus, $a < b \implies b = a + d$ for some positive number d .

Now consider the converse. From Definition 2.2.11, $b = a + d$ implies $b \geq a$, or $a \leq b$. If $b = a$, by Proposition 2.2.6 (cancellation law), $d = 0$, which contradicts the given condition that d is positive. Therefore, $a \neq b$, which implies $a < b$.

Because the statement and the converse are both true, we can conclude that for some positive number d , $a < b$ if and only if $b = a + d$. $\square$

Exercise 2.2.4. *Justify the three lemmas marked with (why?) in Proposition 2.2.13 (trichotomy of order for natural numbers).*

Solution.

(a) For all b , $0 \leq b$.

From the definition of addition, $0 + b = b$. By Definition 2.2.11, this implies $0 \leq b$. $\square$

(b) If $a > b$, then $a++ > b$.

From Definition 2.2.11, $a = b + m$, where m is a natural number. Thus, by the definition of addition, $a++ = (b + m)++ = b + m++$. Because $m++$ is positive by Proposition 2.2.8, $a++ > b$. $\square$

(c) If $a = b$, then $a++ > b$.

From Axiom 2.4, if $a = b$, then $a++ = b++$. Recall that $b++ = b + 1$, thus $a++ = b + 1$. By Definition 2.2.11, this implies $a++ > b$. $\square$

Exercise 2.2.5. Prove Proposition 2.14 (Strong principle of induction): Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m . Suppose that for each $m \geq m_0$, we have the following implication: if $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. Then we can conclude that $P(m)$ is true for all natural numbers $m \geq m_0$.

Solution. Let $Q(n)$ be the property such that $P(m)$ is true for all $m_0 \leq m < n$. Note that the supposed implication becomes “If $Q(n)$ is true, then $P(n)$ is also true. We induct on n .

First, consider the base case $n \leq m_0$. In this case, $Q(n)$ is vacuously true as there is no such natural number m . Hence, the base case is proven.

Now, we assume inductively that $Q(n)$ is true; we will now prove $Q(n++)$ is true. By the inductive hypothesis, $P(m)$ is true for all m where $m_0 \leq m < n$. According to the supposed implication, because $Q(n)$ is true, $P(n)$ is also true. Therefore, $P(m)$ is true for $m_0 \leq m \leq n$, which implies $Q(n++)$ is also true. This closes the induction. $\square$

Exercise 2.2.6. Let n be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m++)$ is true, then $P(m)$ is true. Suppose that $P(n)$ is also true. Prove that $P(m)$ is true for all natural numbers $m \leq n$; this is known as the principle of backwards induction.

Solution. Write $Q(n)$ as the property “If $P(n)$ is true, $P(m)$ is true for all natural numbers $m \leq n$. We apply induction on n .

Consider the base case $n = 0$. The statement $m \leq n$ implies $0 = m + x$ for a natural number x . According to Corollary 2.2.9, this leads to $m = x = 0$. Thus,

$Q(n)$ is equivalent to “If $P(n)$ is true, $P(n)$ is true.”, which is always true regardless of whether $P(n)$ is true or false. Hence, the base case is proven.

Suppose inductively that $Q(n)$ is true; we now need to show $Q(n++)$ is true. From the definition of $P(m)$, $P(n++)$ being true implies $P(n)$ is true. By the inductive statement, $P(m)$ is true for all natural numbers $m \leq n$. Thus if $P(n++)$ is true, $P(m)$ is true for all $m \leq n++$. This is precisely the definition of $Q(n++)$, so $Q(n++)$ is also true. This closes the induction. $\square$

2.3 Multiplication

Exercise 2.3.1. *Prove Lemma 2.3.2 (Commutativity of multiplication): Let n, m be natural numbers. Then $n \times m = m \times n$.*

Solution. First, we introduce a few lemmas. For all lemmas, we induct on n , keeping m fixed (when it appears).

Lemma. *For any natural number n , $n \times 0 = 0$.*

Proof. In the base case $n = 0$, $0 \times 0 = 0$ is true by the definition of multiplication, and 0 is a natural number. Now suppose inductively that $n \times 0 = 0$; we will now prove $(n++) \times 0 = 0$. By the definition of multiplication, $(n++) \times 0 = (n \times 0) + 0 = 0 + 0 = 0$. $\square$

Lemma. *For any natural numbers n and m , $n \times (m++) = (n \times m) + n$.*

Proof. In the base case $n = 0$, $0 \times m++ = 0$ by the definition of multiplication. Also, $(0 \times m) + 0 = 0$. Thus, the left-hand side is true.

Now suppose inductively that $n \times (m++) = (n \times m) + n$; we must now show $(n++) \times (m++) = ((n++) \times m) + (n++)$. We start on the left-hand side.

$$\begin{aligned}
 & (n++) \times (m++) \\
 &= n \times (m++) + m++ && \text{(Definition 2.3.1)} \\
 &= (n \times m) + n + m++ && \text{(Inductive hypothesis)} \\
 &= (n \times m) + m + n++ && \text{(Addition properties)} \\
 &= ((n++) \times m) + (n++) && \text{(Definition 2.3.1)}
 \end{aligned}$$

This closes the induction. □

We can now prove the main statement $n \times m = m \times n$. Consider the base case $n = 0$. Thus, $0 \times m = m \times 0 = 0$. Hence, the base case is true.

Now suppose inductively that $n \times m = m \times n$; we will now prove $(n++) \times m = m \times (n++)$. By the definition of multiplication, $(n++) \times m = (n \times m) + m$. From the second lemma, the right-hand side is also equivalent to $(n \times m) + m$. Because both sides are equal, the statement is true. This closes the induction. □

Exercise 2.3.2. *Prove Lemma 2.3.3 (Positive natural numbers have no zero divisors): Let m, n be natural numbers. Then $n \times m = 0$ if and only if at least one of n, m is equal to zero. In particular, if n and m are both positive, then nm is also positive.*

Solution. We will prove the second statement first. For the sake of contradiction, suppose there exist positive numbers m, n such that $mn = 0$. Because m, n are positive numbers, write $m = p++$, $n = q++$, thus $(q++) \times (p++) = 0$. From the definition of multiplication, $(q \times (p++)) + (p++) = 0$. According to Corollary 2.2.9, this implies $q \times (p++) = p++ = 0$, which is contradictory as Axiom 2.3 states 0 is not the successor of any natural number. Hence, if m, n are both positive, then mn is positive.

Now consider the first statement. The left-to-right implication is simply the contrapositive of the second statement, which is true. Observe the right-to-left implication. If $n = 0$, then $0 \times m = 0$ by the definition of multiplication. If $m = 0$, then by commutativity of multiplication, $m \times 0 = 0 \times m = 0$. Hence, the statements are proven as desired. □

Exercise 2.3.3. *Prove Proposition 2.3.5 (Associativity of multiplication): For any natural numbers a, b, c , we have $(a \times b) \times c = a \times (b \times c)$.*

Solution. We induct on a , keeping b and c fixed.

First, consider the base case $a = 0$. According to the definition of multiplication, the left-hand side leads to $(0 \times b) \times c = 0 \times c = 0$. Similarly, the right-hand side results in $0 \times (b \times c) = 0$. Because both sides are equal, our base case is proven.

Now we assume inductively that $(a \times b) \times c = a \times (b \times c)$; we need to show that $((a+1) \times b) \times c = (a+1)(b \times c)$. By the distributive law, the left-hand side results in $((a+1) \times b) \times c = (a \times b + b) \times c = (a \times b) \times c + b \times c$. The right-hand side produces $(a+1) \times (b \times c) = a \times (b \times c) + b \times c$, which, by the inductive hypothesis, is equal to $(a \times b) \times c + b \times c$. Thus, both sides are equal, and this closes the induction. $\square$

Exercise 2.3.4. *Prove the identity $(a+b)^2 = a^2 + 2ab + b^2$ for all natural numbers a, b .*

Solution. By Proposition 2.3.4 (distributive law) and Proposition 2.3.5 (commutativity of multiplication):

$$\begin{aligned} (a+b)^2 &= (a+b)(a+b) \\ &= a(a+b) + b(a+b) \\ &= a^2 + ab + ab + b^2 \\ &= a^2 + 2ab + b^2 \end{aligned}$$

Exercise 2.3.5. *Prove Proposition 2.3.9 (Euclidean algorithm): Let n be a natural number, and let q be a positive number. Then there exists natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$.*

Solution. We induct on n , fixing q .

First, observe the base case $n = 0$. There is an obvious solution of $m = 0, r = 0$. Hence, the base case is proven.

Now, suppose inductively that there exist natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$; we now have to show there exists natural numbers m', r' such that $0 \leq r' < q$ and $n+1 = m'q + r'$. By the induction hypothesis, $n+1 = mq + r + 1$. Since $r < q$, $r+1 \leq q$. Thus, we have two cases:

- (a) $r+1 < q$: In this case, we can have $n+1 = mq + (r+1)$ with $0 \leq r+1 < q$. Thus, we can have $m' = m, r' = r+1$.
- (b) $r+1 = q$: In this case, we have $n+1 = mq + q = (m+1)q$ by the distributive law. Therefore, $m' = m+1, r' = 0$.

This closes the induction. $\square$

3 Set theory

3.1 Fundamentals

Exercise 3.1.1. *Show that the definition of equality in Definition 3.1.4 is reflexive, symmetric, and transitive.*

Solution. Let A , B , and C be sets.

(a) *Reflexivity:* $A = A$

For every element x of A , x is clearly an element of A . Thus by Definition 3.1.4, $A = A$.

(b) *Symmetry:* If $A = B$, then $B = A$.

Suppose $A = B$. Then for every element $x \in B$, $x \in A$, and for every element $y \in A$, $y \in B$. Thus $B = A$.

(c) *Transitivity:* If $A = B$ and $B = C$, then $A = C$.

Suppose $A = B$ and $B = C$. Thus, for an arbitrary element $x \in A$, $x \in B$ because $A = B$, and since $B = C$, we see that $x \in C$. Similarly, for an arbitrary element $y \in C$, it follows that $y \in B$, and then $y \in A$. Hence, by Definition 3.1.4, it follows that $A = C$ if $A = B$ and $B = C$. $\square$

Exercise 3.1.2. *Using only Definition 3.1.4, Axiom 3.1, Axiom 3.2, and Axiom 3.3, prove that the sets $\emptyset$, $\{\emptyset\}$, $\{\{\emptyset\}\}$, and $\{\emptyset, \{\emptyset\}\}$ are all distinct (i.e., no two of them are equal to each other).*

Solution. First, let's show that $\emptyset$ is different from every other set. The empty set $\emptyset$ has no elements. However, $\emptyset$ is an element of $\{\emptyset\}$ and $\{\emptyset, \{\emptyset\}\}$, and $\{\emptyset\}$ is an element of $\{\{\emptyset\}\}$. Thus, $\emptyset$ is different from the other sets.

Now observe the set $\{\emptyset\}$. According to Axiom 3.3, $x \in \{\emptyset\}$ if and only if $x = \emptyset$. However, $\{\emptyset\}$ is an element of both $\{\{\emptyset\}\}$ and $\{\emptyset, \{\emptyset\}\}$. Hence, $\{\emptyset\}$ is different from the other sets.

We now need to show that $\{\{\emptyset\}\} \neq \{\emptyset, \{\emptyset\}\}$. Notice $\emptyset \notin \{\{\emptyset\}\}$ but $\emptyset \in$

$\{\emptyset, \{\emptyset\}\}$. Therefore, $\{\{\emptyset\}\}$ and $\{\emptyset, \{\emptyset\}\}$ are distinct sets too. We conclude that the sets $\emptyset$, $\{\emptyset\}$, $\{\{\emptyset\}\}$, and $\{\emptyset, \{\emptyset\}\}$ are all distinct. $\square$

Exercise 3.1.3. *Prove the remaining claims in Lemma 3.1.13.*

Solution.

(a) *If a and b are objects, then $\{a, b\} = \{a\} \cup \{b\}$.*

By Axiom 3.4, for an object x , $x \in \{a\} \cup \{b\} \iff (x \in \{a\} \text{ or } x \in \{b\})$. Also, Axiom 3.3 states that $x \in \{a\} \iff x = a$. Similarly, $x \in \{b\} \iff x = b$. Axiom 3.3 also states that $x = a$ or $x = b$ if and only if $x \in \{a, b\}$. Therefore, if $x \in \{a\} \cup \{b\}$, $x \in \{a, b\}$. According to Definition 3.1.4, this implies $\{a, b\} = \{a\} \cup \{b\}$. $\square$

(b) *(Commutativity of the union operation) $A \cup B = B \cup A$.*

For any object x , $x \in A \cup B \iff (x \in A \text{ or } x \in B)$. If $x \in A$, then by Axiom 3.4, then $x \in B \cup A$. Similarly, if $x \in B$, then $x \in B \cup A$. We can show in a similar fashion that $x \in B \cup A \implies x \in A \cup B$. Hence, by Definition 3.1.4, $A \cup B = B \cup A$. $\square$

(c) $A \cup A = A \cup \emptyset = \emptyset \cup A = A$.

First, by Axiom 3.4, for an object x , $x \in A \cup A \iff (x \in A \text{ or } x \in A)$. But, $(x \in A \text{ or } x \in A)$ is equivalent to $x \in A$. So, $x \in A \cup A \iff x \in A$, thus $A \cup A = A$.

Now consider $A \cup \emptyset$. From Axiom 3.4, for an object y , $y \in A \cup \emptyset \iff (y \in A \text{ or } y \in \emptyset)$. However, Axiom 3.2 states that for an arbitrary y , $y \notin \emptyset$. Therefore, $y \in A \cup \emptyset \iff y \in A$, which implies $A \cup \emptyset = A$.

As we have proven the union operation is commutative, $A \cup \emptyset = \emptyset \cup A = A$. $\square$

Exercise 3.1.4. *Prove the remaining claims in Proposition 3.1.18.*

Solution.

(a) *If $A \subseteq B$ and $B \subseteq A$, then $A = B$.*

According to Definition 3.1.15, because $A \subseteq B$, for any object x , $x \in A \implies x \in B$. Similarly, because $B \subseteq A$, $x \in B \implies x \in A$. Thus, $x \in A \iff x \in B$, which implies $A = B$.

(b) If $A \subsetneq B$ and $B \subsetneq C$, then $A \subsetneq C$.

Let x be an arbitrary element of A . Since $A \subsetneq B$ and $B \subsetneq C$, it follows that $x \in B$, and thus $x \in C$. This shows that $A \subseteq C$. However, because $A \subsetneq B$, there exists an element $y \in B$ with $y \notin A$. Since $B \subsetneq C$, $y \in C$. Thus, we have an element y where $y \in C$ and $y \notin A$. This shows that $A \neq C$. Hence, because $A \subseteq C$ and $A \neq C$, we can conclude that $A \subsetneq C$.

Exercise 3.1.5. Let A, B be sets. Show that the three statements $A \subseteq B$, $A \cup B = B$, $A \cap B = A$ are logically equivalent (any one of them implies the other two).

Solution. First, we show that $A \subseteq B \implies A \cup B = B$. To prove $A \cup B = B$, we need to prove that if $A \subseteq B$, then $x \in A \cup B \iff x \in B$ for any object x . Clearly, $x \in B \implies x \in A \cup B$, so we only need to prove $x \in A \cup B \implies x \in B$. Because $A \subseteq B$, we know $x \in A \implies x \in B$. Thus, $x \in A \cup B$ implies $x \in A$ or $x \in B$, which is logically equivalent to $x \in B$. Therefore, $A \subseteq B \implies A \cup B = B$.

Now consider $A \cup B = B \implies A \cap B = A$. To prove $A \cap B = A$, we need to show that if $A \cup B = B$, then $x \in A \cap B \iff x \in A$. Proving $x \in A \cap B \implies x \in A$ is trivial based on Definition 3.1.23. Consider the converse implication $x \in A \implies x \in A \cap B$. Suppose $x \in A$. This implies $x \in A \cup B$, and because $A \cup B = B$, $x \in B$. By Definition 3.1.23, $x \in A$ and $x \in B \iff x \in A \cap B$. Therefore, $A \cup B = B \implies A \cap B = A$.

Lastly, we need to prove $A \cap B = A \implies A \subseteq B$. Because $A \cap B = A$, for all $x \in A$, $x \in A \cap B$. However, $x \in A \cap B \iff x \in A$ and $x \in B$. Therefore, $x \in A \implies x \in B$. $\square$

Exercise 3.1.6. Prove Proposition 3.1.28 (Sets form a boolean algebra).

Solution. Let A, B, C be sets, and let X be a set containing A, B, C as subsets.

(a) (Minimal element) We have $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.

Consider $A \cup \emptyset$. From Axiom 3.4, for an object y , $y \in A \cup \emptyset \iff (y \in A \text{ or } y \in \emptyset)$. However, Axiom 3.2 states that for an arbitrary y , $y \notin \emptyset$. Therefore, $y \in A \cup \emptyset \iff y \in A$, which implies $A \cup \emptyset = A$. $\square$

Now consider $A \cap \emptyset$. For an object y , $y \in A \cap \emptyset \iff (y \in A \text{ or } y \in \emptyset)$. However, Axiom 3.2 states that for an arbitrary y , $y \notin \emptyset$. Therefore, no elements fulfil such conditions, or $A \cap \emptyset = \emptyset$. $\square$

(b) (*Maximal element*) We have $A \cup X = X$ and $A \cap X = A$.

Consider $A \cup X$. From Axiom 3.4, for an object y , $y \in A \cup X \iff y \in A \text{ or } y \in X$. However, if $y \in A$, then $y \in X$ because $A \subseteq X$. Thus, $y \in A \cup X \iff y \in A$, or $A \cup X = X$.

Now consider $A \cap X = A$. From Definition 3.1.23, $y \in A \cap X \iff y \in A \text{ and } y \in X$. However, if $y \in A$, then $y \in X$ because $A \subseteq X$. Thus, $y \in A \cap X \iff y \in A$, or $A \cap X = A$. $\square$

(c) (*Identiity*) We have $A \cap A = A$ and $A \cup A = A$.

First, consider $A \cap A = A$. Definition 3.1.23 states that if an arbitrary object $y \in A \cap A$, then $y \in A$ and $y \in A$. Thus, $y \in A \cap A \iff y \in A$, or $A \cap A = A$ as required. Similarly, for $A \cup A = A$, Axiom 3.4 states that if an arbitrary object $y \in A \cup A$, then $y \in A$ or $y \in A$. This leads to $y \in A \cup A \iff y \in A$, or $A \cup A = A$ as required. $\square$

(d) (*Commutativity*) We have $A \cup B = B \cup A$ and $A \cap B = B \cap A$.

Commutativity of the union operation, or $A \cup B = B \cup A$ has been proven in Exercise 3.1.3 (b). We will only show that the intersection operation is commutative, or $A \cap B = B \cap A$. Let y be an object such that $y \in A \cap B$. Then $y \in A$ and $y \in B$. This is logically equivalent to $y \in B$ and $y \in A$. The converse is proven in a similar fashion, hence, $y \in A \cap B \iff y \in B \cap A$, or $A \cap B = B \cap A$. $\square$

(e) (*Associativity*) We have $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.

The proof of the associativity of the union operation can be found in Lemma 3.1.13. We will only prove the associativity of the intersection operation. By

Definition 3.1.14, we need to prove that all elements of $(A \cap B) \cap C$ is an element of $A \cap (B \cap C)$. Suppose y is an arbitrary element of $(A \cap B) \cap C$. Thus, by Definition 3.1.23, y is both an element of $(A \cap B)$ and C . Because $y \in A \cap B$, $y \in A$ and $y \in B$. Since $y \in B$ and $y \in C$, then $y \in B \cap C$. Also, as $y \in A$, we can conclude that $y \in A \cap (B \cap C)$. The converse is proven similarly. Hence, $(A \cap B) \cap C = A \cap (B \cap C)$ as desired. $\square$

- (f) (*Distributivity*) We have $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

First consider $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. Suppose $x \in A \cap (B \cup C)$. Then, $x \in A$ and $x \in B \cup C$. So $x \in B$ or $x \in C$. We will now split into two cases. If $x \in B$, then $x \in A \cap B$, thus $x \in (A \cap B) \cup (A \cap C)$. Similarly, if $x \in C$, then $x \in A \cap C$, thus $x \in (A \cap B) \cup (A \cap C)$. Thus, $x \in A \cap (B \cup C) \implies x \in (A \cap B) \cup (A \cap C)$.

For the converse, suppose instead $x \in (A \cap B) \cup (A \cap C)$. Then, $x \in A \cap B$ or $x \in A \cap C$. If $x \in A \cap B$, then $x \in A$ and $x \in B$. Because $x \in B$, we know $x \in B \cup C$. Therefore, $x \in A \cap (B \cup C)$. In a similar fashion, if $x \in A \cap C$, we can conclude that $x \in A \cap (B \cup C)$. Hence, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ as required.

We will now consider $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. Suppose x is an element of $A \cup (B \cap C)$. We know that $x \in A$ or $x \in B \cap C$. We will now split into two cases. If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$. If instead $x \in B \cap C$, then $x \in B$ and $x \in C$, so $x \in A \cup B$ and $x \in A \cup C$ as well. In both cases, we can conclude that $x \in (A \cup B) \cap (A \cup C)$. $\square$

- (g) (*Partition*) We have $A \cup (X \setminus A) = X$ and $A \cap (X \setminus A) = \emptyset$.

We will first consider $A \cup (X \setminus A)$. We know that $x \in A$ or $x \in X \setminus A$. If $x \in A$, then $x \in X$ because $A \subseteq X$. If $x \in X \setminus A$ instead, then $x \in X$ and $x \notin A$. Either way, we obtain $x \in X$, thus $x \in A \cup (X \setminus A) \implies x \in X$.

Now consider the converse. Suppose instead $x \in X$. If $x \in A$, then $x \in A \cup (X \setminus A)$. If instead $x \notin A$, then $x \in X \setminus A$ because $x \in X$. Either way, $x \in X \implies x \in A \cup (X \setminus A)$. Hence, $A \cup (X \setminus A) = X$ as desired.

Consider instead $A \cap (X \setminus A)$. Suppose $x \in A \cap (X \setminus A)$. Then, $x \in A$ and

$x \in X \setminus A$. Because $x \in X \setminus A$, $x \notin A$, which contradicts $x \in A$ obtained earlier. Thus, no element x fulfills $x \in A \cap (X \setminus A)$, or $x \in \emptyset$. Hence $A \cap (X \setminus A) = \emptyset$. $\square$

- (h) (*De Morgan Laws*) We have $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ and $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$.

First prove $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$. Suppose $x \in X \setminus (A \cup B)$. Then, $x \in X$ and $x \notin A \cup B$, or $x \notin A$ and $x \notin B$. Thus, we obtain $x \in X \setminus A$ and $x \in X \setminus B$. This leads to $x \in (X \setminus A) \cap (X \setminus B)$.

Now consider the converse. Suppose instead $x \in (X \setminus A) \cap (X \setminus B)$. Because $x \in X \setminus A$ and $X \setminus B$, $x \in X$, $x \notin A$ and $x \notin B$. Thus, $x \notin A \cup B$, and $x \in X \setminus (A \cup B)$. Hence, $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$.

We will now show that $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$. Suppose $x \in X \setminus (A \cap B)$. Then, $x \in X$ and $x \notin A \cap B$. Because $x \notin A \cap B$, $x \notin A$ or $x \notin B$. If $x \notin A$, then $x \notin A$ or $x \notin B$ since the former statement is true. Thus, $x \in (X \setminus A)$ or $x \in (X \setminus B)$, or $x \in (X \setminus A) \cup (X \setminus B)$.

Consider instead the converse. Suppose instead $x \in (X \setminus A) \cup (X \setminus B)$. Then, $x \in (X \setminus A)$ or $x \in (X \setminus B)$. If $x \in (X \setminus A)$, then $x \in X$ and $x \notin A$. Because $x \notin A$, $x \notin A \cap B$. Thus, $x \in X \setminus (A \cap B)$ because $x \in X$. Similarly, if $x \in (X \setminus B)$, $x \in X \setminus (A \cap B)$. Hence, we obtain $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$. $\square$

Exercise 3.1.7. Let A, B, C be sets. Show that $A \cap B \subseteq A$ and $A \cap B \subseteq B$. Furthermore, show that $C \subseteq A$ and $C \subseteq B$ if and only if $C \subseteq A \cap B$. In a similar spirit, show that $A \subseteq A \cup B$ and $B \subseteq A \cup B$, and furthermore that $A \subseteq C$ and $B \subseteq C$ if and only if $A \cup B \subseteq C$.

Solution.

- (a) Show that $A \cap B \subseteq A$ and $A \cup B \subseteq B$.

From Definition 3.1.23, for all objects $x \in A \cap B$, $x \in A$ and $x \in B$. Thus, $x \in A \cap B \implies x \in A$, which is equivalent to $A \cap B \subseteq A$. Similarly, using the commutativity of the union operation, $A \cup B \subseteq B$.

- (b) Show that $C \subseteq A$ and $C \subseteq B$ if and only if $C \subseteq A \cap B$.

Suppose $x \in C$. We will first prove the implication. If $C \subseteq A$ and $C \subseteq B$, then $x \in A$ and $x \in B$. Therefore, $x \in A \cap B$, and $C \subseteq A \cap B$. Now consider the converse. If $C \subseteq A \cap B$, then $x \in A \cap B$. Thus, $x \in A$ and $x \in B$, or $C \subseteq A$ and $C \subseteq B$.

(c) *Show that $A \subseteq A \cup B$ and $B \subseteq A \cup B$.*

For all objects $x \in A$, $x \in A \cup B$, thus $A \subseteq A \cup B$. Similarly, for all objects $x \in B$, $x \in A \cup B$, thus $B \subseteq A \cup B$.

(d) *Show that $A \subseteq C$ and $B \subseteq C$ if and only if $A \cup B \subseteq C$.*

We will first prove the implication. Suppose $x \in A \cup B$. Then, $x \in A$ or $x \in B$. If $x \in A$, then $x \in C$ because $A \subseteq C$. Similarly, if $x \in B$, then $x \in C$. Therefore, $A \cup B \subseteq C$.

Now consider the converse. Suppose instead $x \in A$, thus $x \in A \cup B$. Because $A \cup B \subseteq C$, $x \in A \cup B \implies x \in C$. Thus $x \in A \implies x \in C$, or $A \subseteq C$. Similarly, by swapping the roles of A and B in the above argument, $B \subseteq C$. Hence, the statement is proven. $\square$

Exercise 3.1.8. *Let A, B be sets. Prove the absorption laws $A \cap (A \cup B) = A$ and $A \cup (A \cap B) = A$.*

Solution. First, consider $A \cap (A \cup B) = A$. Suppose x is an element of $A \cap (A \cup B)$, thus $x \in A$ and $x \in A \cup B$. In particular, because $x \in A$, we obtain $A \cap (A \cup B) \subseteq A$. Instead, let $x \in A$, thus $x \in A \cup B$. This leads to $x \in A \cap (A \cup B)$, so $A \subseteq A \cap (A \cup B)$. Hence, $A \cap (A \cup B) = A$.

Now consider $A \cup (A \cap B) = A$. Using the distributivity of the union operation proven in Proposition 3.1.28 (f), the left-hand side is equivalent to $(A \cup A) \cap (A \cup B) = A \cap (A \cup B)$. From the previous results, this is equal to A . Therefore, we obtain $A \cup (A \cap B) = A$ as desired. $\square$

Exercise 3.1.9. *Let A, B, X be sets such that $A \cup B = X$ and $A \cap B = \emptyset$. Show that $A = X \setminus B$ and $B = X \setminus A$.*

Solution. Suppose $x \in A$. From $A \cup B = X$, we know $x \in A \cup B$, and thus $x \in X$.

Also, from $A \cap B = \emptyset$, A and B are disjoint, therefore $x \notin B$. Thus, because $x \in X$ and $x \notin B$, $x \in X \setminus B$. Suppose instead that $x \in X \setminus B$. This implies $x \in X$ and $x \notin B$. Because $x \notin B$ and $x \in A \cup B$, $x \in A$. Hence, $A = X \setminus B$. In a similar fashion, we can swap the roles of A and B in the above proof to show $B = X \setminus A$. $\square$

Exercise 3.1.10. *Let A and B be sets. Show that the three sets $A \setminus B$, $A \cap B$ and $B \setminus A$ are disjoint, and that their union is $A \cup B$.*

Solution. To show the three sets are mutually disjoint, we prove that the intersections of any two of the sets have no elements. Elements in $A \setminus B$ are in A and not B . However, elements in $A \cap B$ are in both A and B , and elements in $B \setminus A$ are in set B and not A . Because an object cannot both be in a set and not in a set, all three intersections are empty sets.

We will now prove that the union of the three sets is $A \cup B$. Consider the following cases for an object x :

- (a) $x \in A, x \in B$: $x \in A \cap B$;
- (b) $x \in A, x \notin B$: $x \in A \setminus B$;
- (c) $x \notin A, x \in B$: $x \in B \setminus A$,
- (d) $x \notin A, x \notin B$: x is not an element of any of the sets, and is therefore also not in the union.

Thus, if $x \in A$ or $x \in B$, x is an element of exactly one of the sets. This is logically equivalent to saying $A \cup B$ is the union of the three sets as per Axiom 3.4. $\square$

Exercise 3.1.11. *Show that the axiom of replacement implies the axiom of specification.*

Solution. Let A be a set, and $P(x)$ be a property pertaining to x . To prove the axiom of specification, we need to construct the set $\{x \in A : P(x)\}$ using only the axiom of replacement. Let $Q(x, y)$ be the property that $y = x$ and $P(x)$ is true. To use the axiom of replacement, for each $x \in A$, there must be at most one y such that $Q(x, y)$ is true. Indeed, for each $x \in A$, there is only one y such that $y = x$. Therefore, there is at most one y such that $Q(x, y)$ is true. We can now show that

there exists a set $\{y : Q(x, y) \text{ for some } x \in A\}$ such that for an object z ,

$$\begin{aligned} & z \in \{y : Q(x, y) \text{ for some } x \in A\} \\ \iff & Q(x, z) \text{ is true for some } x \in A \\ \iff & x = z \text{ and } P(x) \text{ is true for some } x \in A \\ \iff & P(z) \text{ for some } x \in A. \end{aligned}$$

Thus, we can construct the set $\{x \in A : P(x)\}$, which means we have proven the axiom of specification. $\square$